Photosynthesis

Unit 3C Slides 1 through 13: Honors High School Biology

Slide 1: Photosynthesis

We have already studied how cells release usable energy from glucose through cellular respiration. We are now shifting to the process that introduces much of that chemical energy into living systems. Photosynthesis captures light energy and stores part of it in organic molecules.

As we move through this unit, keep track of two things: the movement of matter, especially carbon, oxygen, and hydrogen, and the transformation of energy from light into chemical forms that cells can use.

Slide 2: How Organisms Obtain Energy

Autotrophs, also called producers, build organic molecules from inorganic materials. Plants are the example we use most often, although algae and some bacteria also perform photosynthesis. They do not create energy. They transform light energy into chemical energy stored in molecules such as glucose.

Heterotrophs, or consumers, obtain organic molecules by consuming other organisms or their products. Both groups still use cellular respiration to transfer energy from food molecules into ATP. The difference is how they obtain the organic molecules that enter respiration.

Slide 3: Plant Anatomy Basics

Plant structures coordinate the materials needed for photosynthesis. Roots absorb water and mineral nutrients from the soil and may store sugars. Stems transport water and dissolved minerals upward through xylem and move sugars through phloem. Cohesion, adhesion, and transpiration pull help move water through the plant.

Leaves provide a large surface area for light capture and gas exchange. Stomata are adjustable pores, usually found in the leaf epidermis, that allow carbon dioxide to enter and oxygen and water vapor to leave. Transpiration is the loss of water vapor through the stomata. Guard cells regulate each opening, helping the plant balance carbon dioxide intake with water conservation.

Slide 4: What Plants Need

Plants need light, carbon dioxide, water, and mineral nutrients. Leaves collect light and take in carbon dioxide through stomata. Roots absorb water and ions from the soil. Chloroplasts contain the pigments and membrane systems that capture light energy and begin converting it into chemical energy.

Carbon, hydrogen, oxygen, nitrogen, phosphorus, sulfur, potassium, magnesium, and other elements become the building materials for carbohydrates, lipids, proteins, and nucleic acids. Glucose is therefore more than an energy source. Its carbon skeletons can contribute to many other organic molecules needed for growth.

Slide 5: Photosynthesis Equation

The simplified equation is six carbon dioxide molecules plus six water molecules, using light energy, to produce one glucose molecule and six oxygen molecules: $6CO_2 + 6H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6O_2$. Chlorophyll captures light but is not consumed.

Compare this equation with cellular respiration. The overall products of one process resemble the reactants of the other, showing how matter cycles between organisms and the environment. The energy relationship differs: photosynthesis stores energy in organic molecules, while cellular respiration releases part of that stored energy and transfers it to ATP.

Slide 6: The Chloroplast

Photosynthesis occurs in chloroplasts. Each chloroplast has an outer and inner membrane. Inside, flattened thylakoids contain chlorophyll, electron carriers, and ATP synthase. Thylakoids form stacks called grana, while the surrounding fluid is the stroma.

The thylakoid membrane supports the light-dependent reactions. The stroma contains enzymes for the Calvin cycle. Like mitochondria, chloroplasts contain their own DNA and ribosomes and reproduce partly independently within the cell. These similarities support the endosymbiotic explanation for their evolutionary origin.

Slide 7: Chloroplast Diagram

Use the diagram to connect each structure with its function. The outer and inner membranes form the chloroplast envelope. The intermembrane space lies between them. The stroma is the fluid-filled interior. A granum is a stack of thylakoids, while a thylakoid is a single membrane-bound disc. Stroma lamellae connect grana.

Distinguishing between the thylakoid membrane and the stroma will become essential. Light energy is captured, and the thylakoid membrane produces ATP and NADPH. The Calvin cycle uses ATP and NADPH in the stroma to help build carbohydrates.

Slide 8: Photosynthesis Overview

Photosynthesis includes two connected sets of reactions. The light-dependent reactions occur in the thylakoid membrane. They capture light energy, split water, release oxygen, and produce ATP and NADPH.

The Calvin cycle occurs in the stroma. It uses carbon dioxide, along with ATP and NADPH from the light reactions, to produce G3P, a three-carbon sugar that can be used to make glucose and other organic molecules. Calling the Calvin cycle light-independent means it does not directly absorb light. It still depends on products made by the light reactions.

Slide 9: Light Absorption

Visible light is a small part of the electromagnetic spectrum. Light travels in waves, and wavelength determines the color that we perceive. White sunlight contains a mixture of visible wavelengths, not a single color.

Wavelength and energy are inversely related. Shorter wavelengths carry more energy per photon than longer wavelengths. Plants cannot use every wavelength equally because pigment molecules absorb only particular ranges of light.

Slide 10: The Spectrum of Color

The visible spectrum extends roughly from violet (about 380 to 400 nanometers) to red (about 700 to 750 nanometers). Violet and blue light have shorter wavelengths and higher energy per photon. Red light has a longer wavelength and lower energy per photon.

Photosynthesis depends on whether pigments absorb a wavelength, not simply on which color carries the most energy. A wavelength can drive photosynthesis only if a pigment captures it and transfers that energy into the photosynthetic pathway.

Slide 11: Pigments of Photosynthesis

Pigments absorb certain wavelengths and reflect or transmit others. Chlorophyll a is the primary photosynthetic pigment and absorbs strongly in the blue-violet and red portions of the spectrum. Chlorophyll b and carotenoids function as accessory pigments that expand the range of usable light.

When a pigment absorbs a photon, one of its electrons moves to a higher energy state. That excited electron can transfer energy into the light-dependent reactions. The arrangement of pigments in photosystems allows the plant to collect light across a broader range than chlorophyll a could capture alone.

Slide 12: Which Color Is Least Absorbed

The graph compares absorption across the visible spectrum. Chlorophyll a and chlorophyll b absorb strongly in blue and red regions but absorb relatively little green light. Much of the green wavelength is reflected or transmitted, which is why healthy leaves usually appear green.

Carotenoids absorb strongly in parts of the blue and blue-green range. Their absorption pattern complements chlorophyll and also helps protect photosynthetic tissues from excess light energy. When reading an absorption spectrum, higher peaks indicate stronger absorption at that wavelength.

Slide 13: Accessory Pigments and Fall Color

Accessory pigments include chlorophyll b, carotene, and xanthophylls. They capture wavelengths that chlorophyll a absorbs less efficiently and transfer some of that energy to the reaction center. Carotene contributes orange colors, while xanthophylls contribute yellow tones.

During fall, shorter daylight and cooler temperatures reduce chlorophyll production, and existing chlorophyll breaks down. As the green pigment disappears, the already present carotenoids and xanthophylls become visible. Some leaves may also develop red and purple colors when they produce anthocyanins.

Closing Statement

Photosynthesis depends on coordinated plant structures, chloroplast compartments, and pigments with different absorption properties. These foundations set the stage for understanding how light energy drives electron transfer and ATP production during the light-dependent reactions.